

AGE-P

An Evolutionary Platform for the Self-Organization of Smart-Appliance Ensembles

Nature-inspired informatics

Technologies are being developed for optimisation of telecom networks that are inspired from nature

There are several striking examples of intelligent design, inspired by nature. We look at some of these technologies

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Philosophers are right when they say "Progress often occurs at the boundaries between disciplines." Our natural world is full of complexity - from the organisation of the nervous system, to molecular genomics and even coral reef ecologies. The same way our technology, computer technology in particular has been manifestly becoming more and more complex—from systems on

chips to interest-router connectivity, and from software module interactions to new operating systems. So, the natural question arises: Are we exploiting insights from biology to help understand the complex computational systems that we've produced or are we doing everything ground-up?

Throughout history we've built artefacts whose inspiration derives from biology and the natural world. You will naturally argue that computer science and technology haven't been influenced to such a large extent because our computers implement a binary logic that's hard to discern in nature, using a general architecture that's remote from anything that has evolved. However, it isn't hard to recognise the influence of biological processes and methods on computer science.

Here are few examples which will tell you how nature has inspired everyday tech:

- ✔ Velcro — inspired by seeds' clingy burrs
- ✔ Low-friction ship hulls — inspired by shark skin
- ✔ Morphing aircraft wings — inspired by bird wings
- ✔ Temperature-adapting fabric — inspired by pinecones
- ✔ Dirt- and water-resistant paint — inspired by the lotus flower

The Network Paradigm

If we take a deep look at the internet, we will see a striking similarity with nature and genes in particular. A gene network is defined by connections between specific genes, whose outputs are influenced by proteins (expressed by other genes), metabolites, and small signaling molecules. Other networks are layered on top of genes, such as protein interaction networks, signalling networks, and metabolic networks, which themselves underpin arrays of interacting networks that link cells together.

An organism's physiology requires the linking of many gene, protein, metabolic, and cellular networks, with feedback interactions between them. In a very similar fashion, our modern internet is a complex, multilayered, and modular structure. Physical devices provide the foundation, and frame formats determine how packets course through the physical links. The IP packet protocol lies above the link layer and enables communication between various computational networks, creating the experience of a single

Naturally Inspired

